Anatomy and Physiology of auditory and vestibular system

AURICLE OR PINNA

The entire pinna except its lobule and the outer part of external acoustic canal are made up of a framework of a single piece of yellow elastic cartilage covered with skin.

A. EXTERNAL ACOUSTIC (AUDITORY) CANAL

It extends from the bottom of the concha to the tympanic membrane and measures about 24 mm along its posterior wall. The canal is divided into two parts: (i) cartilaginous and (ii) bony.

B. TYMPANIC MEMBRANE OR THE DRUMHEAD

It forms the partition between the external acoustic canal and the middle ear. It is obliquely set and as a result, its posterosuperior part is more lateral than its anteroinferior part. It is 9–10 mm tall, 8–9 mm wide and 0.1 mm thick (Fig. 1).

THE MIDDLE EAR

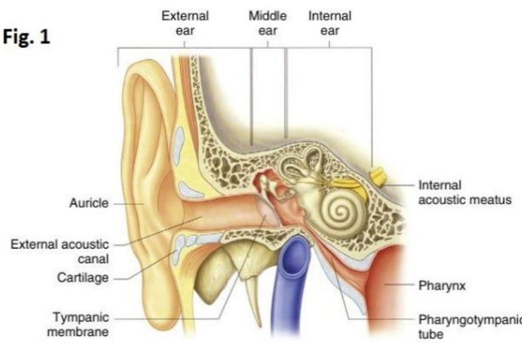
The middle ear together with the eustachian tube, aditus, antrum and mastoid air cells is called middle ear cleft. It is lined by mucous membrane and filled with air. The middle ear extends much beyond the limits of tympanic membrane which forms its lateral boundary and is sometimes divided into:

(i) mesotympanum (lying opposite the pars tensa),

(ii) epitympanum or the attic (lying above the pars tensa but medial to Shrapnell's membrane and the bony lateral attic wall) and

(iii) hypotympanum (lying below the level of pars tensa) (Figure 1.8). The portion of middle ear around the tympanic orifice of the eustachian tube is sometimes called protympanum (Fig. 2).

Fig. 1



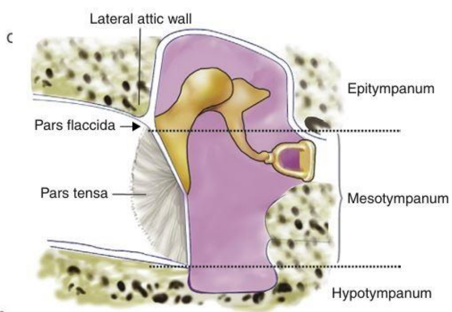


Figure 2 Divisions of middle ear into epi-, meso- and hypo-tympanum.

THE INTERNAL EAR

The internal ear or the labyrinth is an important organ of hearing and balance. It consists of a bony and a membranous labyrinth.

The membranous labyrinth is filled with a clear fluid called **endolymph** while the space between membranous and bony labyrinths is filled with **perilymph**.

BONY LABYRINTH (Fig. 3): It consists of three parts: the vestibule, the semicircular canals and the cochlea.

1. Vestibule. It is the central chamber of the labyrinth. In its lateral wall lies the oval window. The inside of its medial wall presents two recesses, a spherical recess, which lodges the saccule, and an elliptical recess, which lodges the utricle. Below the elliptical recess is the opening of aqueduct of vestibule through which passes the endolymphatic duct. In the posterosuperior part of vestibule are the five openings of semicircular canals.

2. Semicircular canals. They are three in number, the lateral, posterior and superior, and lie in planes at right angles to one another. Each canal has an ampullated end which opens independently into the vestibule and a nonampullated end. The nonampullated ends of posterior and superior canals unite to form a common channel called *crus commune*. Thus, the three canals open into the vestibule by five openings.

3. Cochlea. The bony cochlea is a coiled tube making 2.5 to 2.75 turns round a central pyramid of bone called modiolus. The base of modiolus is directed towards internal acoustic meatus and transmits vessels and nerves to the cochlea. Around the modiolus and winding spirally like the thread of a screw, is a thin plate of

bone called *osseous spiral lamina*. It divides the bony cochlea incompletely and gives attachment to the basilar membrane.
The bony bulge in the medial wall of middle ear, the promontory, is due to the basal coil of the cochlea. The bony cochlea contains three compartments:
(a) Scala vestibuli,
(b) Scala tympani,
(c) Scala media or the membranous cochlea (Fig 4 and 5).
The scala vestibuli and scala tympani are filled with perilymph and communicate with each other at the apex of cochlea through an opening called helicotrema (Fig 5). Scala vestibuli is closed by the footplate of stapes which separates it from the air-filled middle ear. The scala tympani is closed by secondary tympanic membrane; it is also connected with the subarachnoid space through the aqueduct of cochlea.

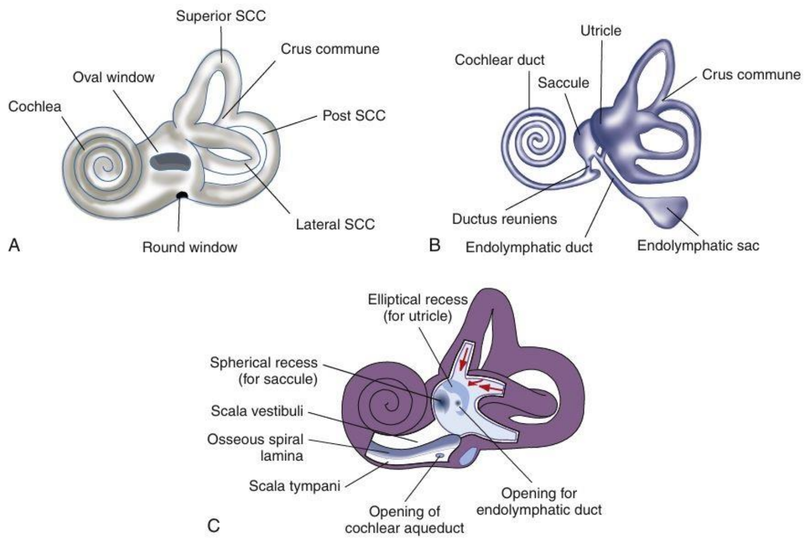


Figure 3 (A) Left bony labyrinth. (B) Left membranous labyrinth. (C) Cut section of bony labyrinth.

MEMBRANOUS LABYRINTH (Fig. 3 and 4) It consists of the cochlear duct, the utricle and saccule, the three semicircular ducts, and the endolymphatic duct and sac.
1. Cochlear duct (Fig 4 and 5). Also called membranous cochlea or the scala media. It is a blind coiled tube. It appears triangular on cross-section and its three walls are formed by:
(a) the basilar membrane, which supports the organ of Corti;
(b) the Reissner's membrane, which separates it from the scala vestibule; and
(c) the stria vascularis, which contains vascular epithelium and is concerned with secretion of endolymph.
Cochlear duct is connected to the saccule by ductus reuniens (Figure 3). The length of basilar membrane increases as we proceed from the basal coil to the apical coil. It is for this reason that higher frequencies of sound are heard at the basal coil while lower ones are heard at the apical coil.
2. Utricle and saccule. The utricle lies in the posterior part of bony vestibule. It receives the five openings of the three semicircular ducts. It is also connected to the saccule through utriculosaccular duct. The sensory epithelium of the utricle is called macula and is concerned with linear acceleration and deceleration. The saccule also lies in the bony vestibule, anterior to the utricle and opposite the stapes footplate. Its sensory epithelium is also called macula. Its exact function is not known. It probably also responds to linear acceleration and deceleration.
3. Semicircular ducts. They are three in number and correspond exactly to the three bony canals. They open in the utricle. The ampullated end of each duct contains a thickened ridge of neuroepithelium called crista ampullaris.
4. Endolymphatic duct and sac. Endolymphatic duct is formed by the union of two ducts, one each from the saccule and the utricle. It passes through the vestibular aqueduct. Its terminal part is dilated to form

endolymphatic sac, which lies between the two layers of dura on the posterior surface of the petrous bone. Endolymphatic sac is surgically important. It is exposed for drainage or shunt operation in Ménière's disease.

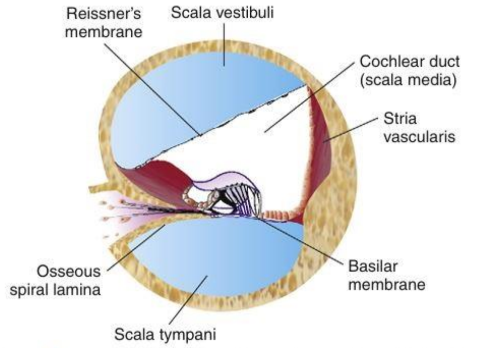


Figure 4 Section through cochlea to show scala media (cochlear duct), scala vestibuli and scala tympani.

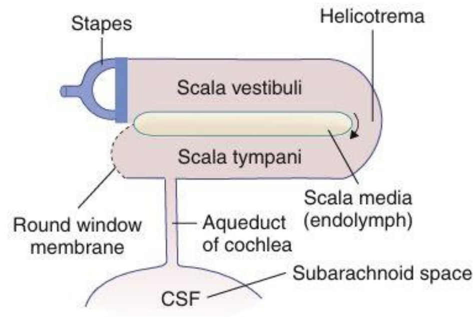


Figure 5 Diagrammatic representation of perilymphatic system. CSF passes into scala tympani through aqueduct of cochlea.

Peripheral Receptors and Physiology of Auditory and Vestibular Systems

ORGAN OF CORTI (Fig. 6) Organ of Corti is the sense organ of hearing and is situated on the basilar membrane. Important components of the organ of Corti are:

1. Tunnel of Corti, which is formed by the inner and outer rods. It contains a fluid called cortilymph. The exact function of the rods and cortilymph is not known.
2. Hair cells. They are important receptor cells of hearing and transduce sound energy into electrical energy. Inner hair cells form a single row while outer hair cells are arranged in three or four rows. Inner hair cells are richly supplied by afferent cochlear fibres and are probably more important in the transmission of auditory impulses. Outer hair cells mainly receive efferent innervation from the olivary complex and are concerned with modulating the function of inner hair cells. Differences between inner and outer hair cells are given in Table 1.
3. Supporting cell. Deiters' cells are situated between the outer hair cells and provide support to the latter. Cells of Hensen lie outside the Deiters' cells.
4. Tectorial membrane. It consists of gelatinous matrix with delicate fibres. It overlies the organ of Corti. The shearing force between the hair cells and tectorial membrane pro duces the stimulus to hair cells.

AUDITORY NEURAL PATHWAYS AND THEIR NUCLEI (Fig 7)

Hair cells are innervated by dendrites of bipolar cells of spiral ganglion which is situated in Rosenthal's canal (canal running along the osseous spiral lamina). Axons of these bipolar cells form the cochlear division of CN VIII and end in the cochlear nuclei, the dorsal and ventral, on each side of the medulla. Further course of auditory pathways is complex. From cochlear nuclei, the main nuclei in

1. Superior olivary complex
2. Nucleus of lateral lemniscus
3. Inferior colliculus
4. Medial geniculate body
5. Auditory cortex

The auditory fibres travel via the ipsilateral and contra lateral routes and have multiple decussation points. Thus each ear is represented in both cerebral hemispheres. The area of cortex, concerned with hearing is situated in the superior temporal gyrus. For auditory pathways, remember the mnemonic E.COLI-MA: Eighth nerve, Cochlear nuclei, Olivary complex, Lateral lemniscus, Inferior colliculus, Medial geniculate body and Auditory cortex.

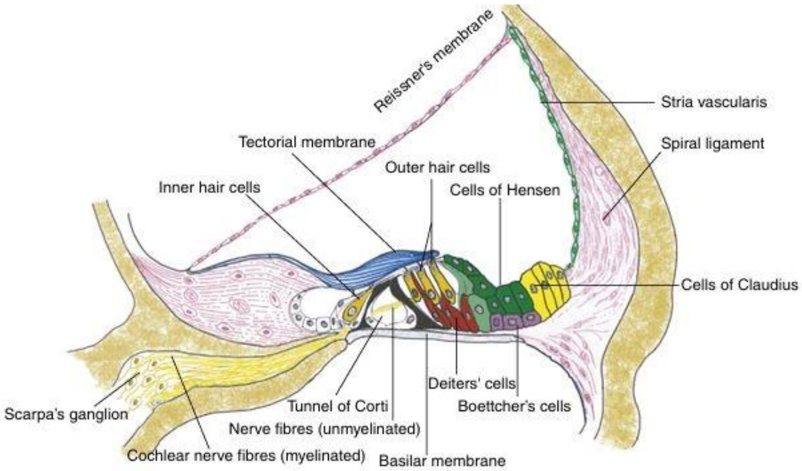


Figure 6 Structure of organ of Corti.

Table 1 Differences between inner and outer hair cells

	Inner hair cells	Outer hair cells
Total no.	3500	12,000
Rows	One row	Three or four rows
Shape	Flask shaped	Cylindrical
Nerve supply	Primarily afferent fibres and very few efferent	Mainly efferent fibres and very few afferent
Development	Develop earlier	Develop late
Function	Transmit auditory stimuli	Modulate function of inner hair cells
Vulnerability	More resistant	Easily damaged by ototoxic drugs and high intensity noise

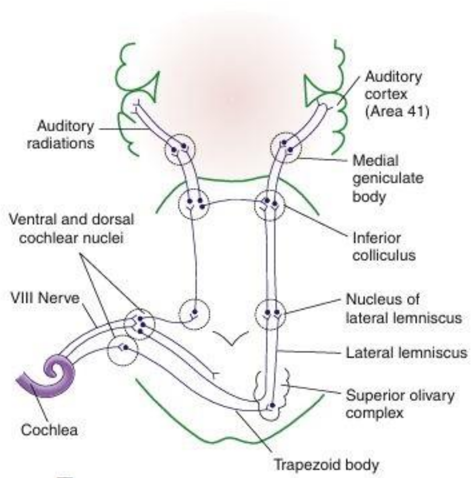


Figure 7 Auditory pathways from the right cochlea. Note bilateral route through brainstem and bilateral cortical representation.

MECHANISM OF HEARING

A sound signal in the environment is collected by the pinna, passes through external auditory canal and strikes the tympanic membrane. Vibrations of the tympanic membrane are transmitted to stapes footplate through a chain of ossicles coupled to the tympanic membrane. Movements of stapes footplate cause pressure changes in the labyrinthine fluids, which move the basilar membrane. This stimulates the hair cells of the organ of Corti. It is these hair cells which act as transducers and convert the mechanical energy into electrical impulses, which travel along the auditory nerve. Thus, the mechanism of hearing can be broadly divided into:

1. Mechanical conduction of sound (conductive apparatus).
2. Transduction of mechanical energy to electrical impulses (sensory system of cochlea).
3. Conduction of electrical impulses to the brain (neural pathways).

1. CONDUCTION OF SOUND

Develop late Modulate function of inner hair cells Easily damaged by ototoxic drugs and high intensity noise A person under water cannot hear any sound made in the air because 99.9% of the sound energy is reflected away from the surface of water because of the impedance offered by it. A similar situation exists in the ear when air-conducted sound has to travel to cochlear fluids. Nature has compensated for this loss of sound energy by interposing the middle ear which converts sound of greater amplitude but lesser force, to that of lesser amplitude but greater force. This function of the middle ear is called impedance matching mechanism or the transformer action. It is accomplished by:

(a) Lever action of the ossicles. Handle of malleus is 1.3 times longer than long process of the incus, providing a mechanical advantage of 1.3.

(b) Hydraulic action of tympanic membrane. The area of tym panic membrane is much larger than the area of stapes footplate, the average ratio between the two being 21:1. As the effective vibratory area of tympanic membrane is only two-thirds, the effective areal ratio is reduced to 14:1, and this is the mechanical advantage provided by the tympanic membrane (Fig. 8).
(c) Curved membrane effect. Movements of tympanic mem brane are more at the periphery than at the centre where malleus handle is attached. This too provides some leverage.

Natural resonance of external and middle ear.
Inherent anatomic and physiologic properties of the external and middle ear allow certain frequencies of sound to pass more easily to the inner ear due to their natural resonances. Natural resonance of external ear canal is 3000 Hz and that of middle ear 800 Hz. Frequencies most efficiently transmitted by ossicular chain are between 500 and 2000 Hz while that by tympanic membrane is 800–1600 Hz. Thus greatest sensitivity of the sound transmission is between 500 and 3000 Hz and these are the frequencies most important to man in day to-day conversation (Table 2).

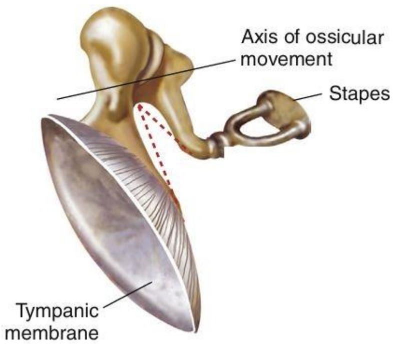


Figure 8 Transformer action of the middle ear. Hydraulic effect of tympanic membrane and lever action of ossicles combine to compensate the sound energy lost during its transmission from air to liquid medium.

Table 2. Natural resonance and efficiency of auditory apparatus

External auditory canal	3000 Hz
Tympanic membrane	800–1600 Hz
Middle ear	800 Hz
Ossicular chain	500–2000 Hz

VESTIBULAR SYSTEM PERIPHERAL RECEPTORS
They are of two types:
1. CRISTAE (Fig. 9) They are located in the ampullated ends of the three semicircular ducts. These receptors respond to angular acceleration.
2. MACULAE (Fig 10) They are located in otolith organs (i.e. utricle and saccule). Macula of the utricle lies in its floor in a horizontal plane. Macula of the saccule lies in its medial wall in a vertical plane. They sense position of head in response to gravity and linear acceleration.

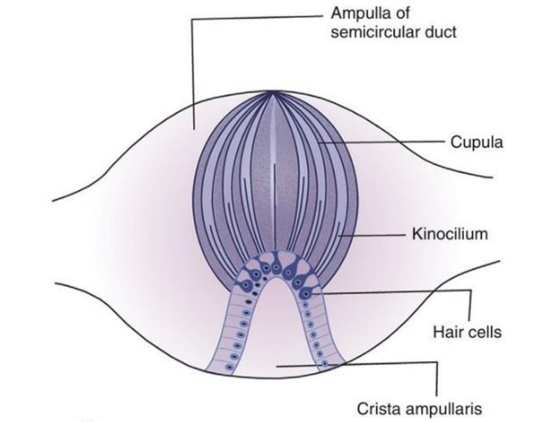


Figure 9 Structure of ampullary end of semicircular duct. Over the crista lie sensory hair cells interspersed with supporting cells. Hair from sensory cells project into the gelatinous substance of cupula.

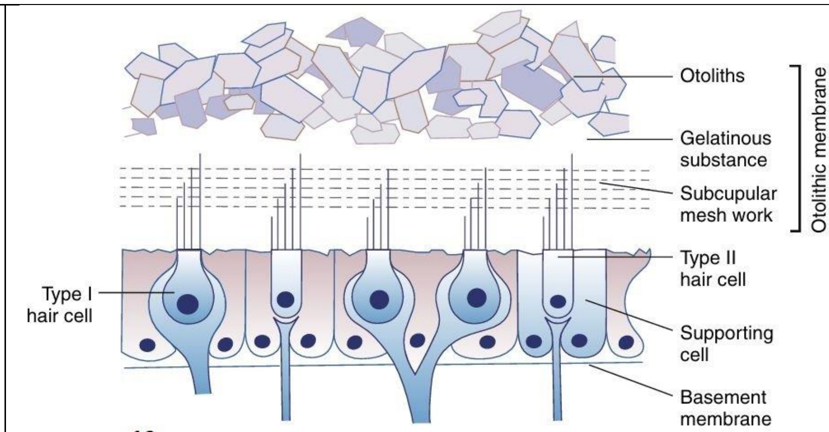


Figure 10 Structure of macula, the sensory end organ of the utricle and the saccule.

VESTIBULAR NERVE Vestibular or Scarpa's ganglion is situated in the lateral part of the internal acoustic meatus. It contains bipolar cells. The distal processes of bipolar cells innervate the sensory epithelium of the labyrinth while its central processes aggregate to form the vestibular nerve. epithelium of the labyrinth while its central processes aggregate to form the vestibular nerve.

CENTRAL VESTIBULAR CONNECTIONS
The fibres of vestibular nerve end in vestibular nuclei and some go to the cerebellum directly. Vestibular nuclei are four in number, the superior, medial, lateral and descending. Afferents to these nuclei come from:
1. Peripheral vestibular receptors (semicircular canals, utri cle and saccule)
2. Cerebellum
3. Reticular formation
4. Spinal cord
5. Contralateral vestibular nuclei
Thus, information received from the labyrinthine receptors is integrated with information from other somatosensory systems. Efferents from vestibular nuclei go to:
1. Nuclei of CN III, IV, VI via medial longitudinal bundle. It is the pathway for vestibulo-ocular reflexes and this explains the genesis of nystagmus.
2. Motor part of spinal cord (vestibulospinal fibres). This coordinates the movements of head, neck and body in the maintenance of balance.
3. Cerebellum (vestibulocerebellar fibres). It helps to coordinate input information to maintain the body balance.
4. Autonomic nervous system. This explains nausea, vomiting, palpitation, sweating and pallor seen in vestibular disorders (e.g. Ménière's disease).
5. Vestibular nuclei of the opposite side.
6. Cerebral cortex (temporal lobe). This is responsible for subjective awareness of motion.

PHYSIOLOGY OF VESTIBULAR SYSTEM
Vestibular system is conveniently divided into:
1. Peripheral, which is made up of membranous labyrinth (semicircular ducts, utricle and saccule) and vestibular nerve. Right SCC Left SCC Figure 11 Rotation test. At the end of rotation to the right, semicircular canals stop but endolymph continues to move to the right, i.e. towards the left ampulla but away from the right, causing nystagmus to the left.
2. Central, which is made up of nuclei and fibre tracts in the central nervous system to integrate vestibular impulses with other systems to maintain body balance.

SEMICIRCULAR CANALS

They respond to angular acceleration and deceleration. The three canals lie at right angles to each other but the one which lies at right angles to the axis of rotation is stimulated the most. Thus horizontal canal will respond maximum to rotation on the vertical axis and so on. Due to this arrangement of the three canals in three different planes, any change in position of head can be detected. Stimulation of semicircular canals produces nystagmus and the direction of nystagmus is determined by the plane of the canal being stimulated. Thus, nystagmus is horizontal from horizontal canal, rotatory from the superior canal and vertical from the posterior canal.

The stimulus to semicircular canal is flow of endolymph which displaces the cupula. The flow may be towards the cupula (ampullopetal) or away from it (ampullofugal), better called utriculopetal and utriculofugal. Ampullopetal flow is more effective than ampullofugal for the horizontal canal. The quick component of nystagmus is always opposite to the direction of flow of endolymph. Thus, if a person is rotated to the right for sometime and then abruptly stopped, the endolymph continues to move to the right due to inertia (i.e. ampullopetal for left canal), the nystagmus will be horizontal and directed to the left (Figure 11). Remember nystagmus is in the direction opposite to the direction of flow of endolymph.

UTRICLE AND SACCULE

Utricle is stimulated by linear acceleration and deceleration or gravitational pull during the head tilts. The sensory hair cells of the macula lie in different planes and are stimulated by displacement of otolithic membrane during the head tilts. The function of saccule is similar to that of utricle as the structure of maculae in the two organs is similar but experimentally, the saccule is also seen to respond to sound vibrations. The vestibular system thus registers changes in the head position, linear or angular acceleration and deceleration, and gravitational effects. This information is sent to the central nervous system where information from other systems— visual, auditory, somatosensory (muscles, joints, tendons, skin)—is also received. All this information is integrated and used in the regulation of equilibrium and body posture. Cerebellum, which is also connected to vestibular end organs, further coordinates muscle movements in their rate, range, force and duration and thus helps in the maintenance of balance.

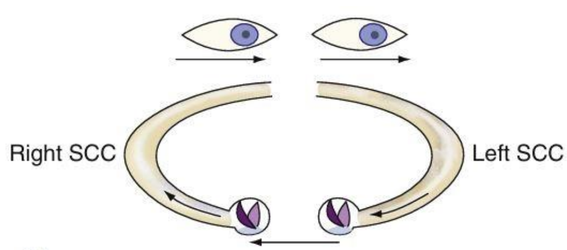


Figure 11 Rotation test. At the end of rotation to the right, semicircular canals stop but endolymph continues to move to the right, i.e. towards the left ampulla but away from the right, causing nystagmus to the left.

Good Luck

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